Emir Muhammet Aran

Medical AI Developer | Computer Engineering Student

Medical AI researcher focused on cell imaging and biological aging simulation. Building generative models for label-free microscopy analysis with clinical deployment potential.

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Medium
Ankara, Turkey

Education

Gazi University 2025 - 2027

B.S. in Computer Engineering

Transferred to Gazi University (merit-based, 50% scholarship at prior institution) - GPA 3.52

Ankara Medipol University 2022 - 2025

B.S. in Computer Engineering

Sophomore: GPA 3.70 | Freshman: GPA 3.68 (Ranked 2nd)

Work Experience

Software Developer Intern Summer 2024

Gibrin R&D | TOBB Garaj

- Built real-time IoT monitoring dashboard integrating ESP32 sensors with Firebase cloud database
- Developed secure authentication system with role-based access control across 3 mobile applications
- Implemented cross-platform Flutter applications; shipped to internal users at TOBB Garaj

Medical AI Projects

BraTS 2020 Brain Tumor Segmentation

3D medical image segmentation using U-Net with EfficientNetB0 achieving 90.34% Sensitivity and 99.96% Specificity for precise tumor region identification (Necrotic, Enhancing Tumor, Edema). Implements Focal Loss for class imbalance with interactive Gradio interface.

U-Net | 90.34% Sensitivity | 3D Segmentation | Focal Loss | ✓ DEPLOYED

Live Demo | GitHub →

Brain Tumor MRI Classification System

Multi-class deep learning classifier achieving 99% accuracy for brain tumor detection (Glioma, Meningioma, Pituitary, No Tumor). Transfer learning with EfficientNetB3 featuring Grad-CAM visualizations for clinical interpretability.

EfficientNetB3 | 99% Acc | Transfer | Grad-CAM | ✓ DEPLOYED

Live Demo | GitHub →

Breast Cancer Histopathology Detection

Custom lightweight CNN from scratch achieving 91.5% sensitivity (exceeds FDA/EMA screening benchmark) and AUC 0.94. Implemented Focal Loss with optimized clinical threshold prioritizing Recall over Accuracy—catching missed diagnoses. Published detailed analysis on blind augmentation pitfalls and architecture design for medical imaging.

Custom CNN | 91.5% Sensitivity | Focal Loss | FDA/EMA Compliant | ✓ DEPLOYED

Live Demo | GitHub → | Article

Chest X-ray Disease Classification

Multi-label deep learning system for 15-class thoracic disease detection from chest X-ray images using EfficientNetB0 with full fine-tuning. Achieves 0.784 mean AUC with Test-Time Augmentation (TTA). Implements Focal Loss for class imbalance, balanced sampling with oversampling for rare diseases, and medical-appropriate threshold optimization for ≥80% recall priority.

EfficientNetB0 | 0.784 AUC | Multi-Label (15 diseases) | Focal Loss | TTA | ✓ OPEN SOURCE

Live Demo | GitHub → | Article

MSC Aging Simulation via Label-Free Microscopy

Generative model for biological age reversal in label-free brightfield microscopy images of Mesenchymal Stem Cells. LDM + LoRA architecture performing img2img translation from senescent to rejuvenated cell morphology.

PyTorch | LDM + LoRA | Bidirectional | img2img | Ongoing Research

Article

Additional Technical Projects

Clinical NLP & Sequence Architectures

Unstructured clinical text structuring (BioBERT) and automatic subtitle translation (Transformer seq2seq, attention mechanisms)

Computer Vision & Infrastructure

High-throughput DICOM image processing pipelines and real-time face recognition (OpenCV + MTCNN pipeline)

Predictive Analytics

Ensemble methods (Random Forest, XGBoost) and time series forecasting (FBProphet) for optimization tasks

View all projects on GitHub (46 repositories)

Technical Skills

Programming Languages

Python, C, C#, Bash

AI/ML Frameworks

PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, XGBoost, HuggingFace Transformers, LLMs

Data Science & Analysis

Pandas, NumPy, Matplotlib, Feature Engineering, Vector Embeddings, Statistical Analysis, LLM Fine-tuning

Development & Deployment

HuggingFace Spaces, REST APIs, Git, Linux, Docker, FastAPI, Flask

Achievements & Leadership

- **ACUHIT Healthcare Hackathon 2026** — Sole AI/ML developer; architected full-stack clinical NLP system (Sentence Transformer + TF-IDF ensemble) with Flask API and React frontend
- **Vesuvius Challenge** — Top 39% | 3D volumetric segmentation of ancient carbonized scrolls
- **Ostim Game Jam 2025** - 1st Place Winner | Led team in rapid game prototype development
- **MEDCODES** — Board Member (2024-Present) | Leading technical training sessions for medical engineering students
- **CYBERMEDU** — Board Member (2023-Present) | Organizing cybersecurity and medical technology education initiatives

Languages

English: C1